



Jyothi
Engineering College
(AUTONOMOUS)

Reaccredited with NAAC (Grade A) and
NBA Programmes (CE, CS, EC, EE, ME, MR)
Jyothi Hills, P. O. Vettikkattiri, Cheruthuruthy
Thrissur, Kerala, India, 679531
04884 259000 | info@jecc.ac.in | www.jecc.ac.in

Approved by AICTE & Affiliated to APJ Abdul Kalam Technological University
A Centre of Excellence in Science and Technology by the Catholic Archdiocese of Trichur

CENTRE FOR WATER RESEARCH AND EDUCATION (CWRE)

Jyothi Engineering College, Cheruthuruthy

SUMMER INTERNSHIP PROGRAMME 2026 REPORT

Water Quality Assessment and Sanitation

Conducted for B.Sc. Microbiology Honours Students

Participating Institution

Markaz College of Arts and Science, Athavanad

Programme Dates

7, 8, 9, 10, 11, 13, 16, 17, 18 & 20 April 2026

(10 Days | 60+ Hours)

Prepared and Submitted by:

Prof. M.G. Cyriac

Chief Supervisor, CWRE

Associate Professor, Civil Engineering

Jyothi Engineering College, Cheruthuruthy

Date: April 2026

Signature: _____

Prof. M.G. Cyriac

Foreword by the Chief Supervisor

This report is prepared and submitted by me, Prof. M.G. Cyriac, Chief Supervisor of the Centre for Water Research and Education (CWRE), Jyothi Engineering College, Cheruthuruthy, in my capacity as the principal organiser and supervisor of the internship programme on Water Quality and Sanitation conducted during April 2026.

The internship was hosted by CWRE for twenty-five students of B.Sc. Microbiology Honours from Markaz College of Arts and Science, Athavanad. Over ten working days — spanning April 7 to April 20, 2026 — the students engaged in classroom instruction, laboratory practice, field surveys in their home areas, an industrial visit to the Peechi Dam and Kerala Water Authority facilities, and a Water Quality and Sanitation Awareness Competition. The total instructional and activity time exceeded sixty hours.

The purpose of this report is to provide a comprehensive account of the programme — its planning, daily conduct, activities, participant performance, and outcomes — for the benefit of the institution, the sponsoring college, and any regulatory or accrediting authority that may require documentation of the programme. All observations and assessments recorded here are based on my direct oversight and interaction with the students, faculty, and resource persons throughout the programme.

It gives me great satisfaction to report that the students from Markaz College demonstrated exceptional dedication, curiosity, and discipline throughout the programme. Their conduct was a credit both to themselves and to their institution.

Prof. M.G. Cyriac

B.Sc. Engg. (Civil), M.Tech. (Environmental Engineering), DMP (IIM Bangalore)
Chief Supervisor, CWRE | Associate Professor, Civil Engineering
Jyothi Engineering College, Cheruthuruthy

1. Programme Overview

Detail	Particulars
Programme Title	Internship on Water Quality Assessment and Sanitation
Host Institution	Centre for Water Research and Education (CWRE), Jyothi Engineering College, Cheruthuruthy
Participating Institution	Markaz College of Arts and Science, Athavanad
Programme for	B.Sc. Microbiology Honours (Final Year)
Number of Students	25
Programme Days	7, 8, 9, 10, 11, 13, 16, 17, 18, 20 April 2026 (10 days)
Total Duration	More than 60 hours
Chief Supervisor	Prof. M.G. Cyriac, Associate Professor, Civil Engineering, Jyothi Engineering College
Programme Coordinators	Mrs. Jeffy Johny and Mr. Suhas Nair

1.1 Objectives

The programme was designed around five core objectives:

- To familiarise students with the physical and chemical methods of water quality assessment applicable to drinking water.
- To provide practical exposure to the microbiological analysis of water, including the identification and significance of indicator organisms.
- To sensitise students to the specific water quality challenges prevalent in Kerala, including iron contamination, well pollution, and seasonal bacteriological deterioration.
- To explain and demonstrate both conventional (treatment plant-scale) and domestic-level water and wastewater treatment processes.
- To train students in the application of sanitary measures for the proper upkeep of buildings and their surroundings.

1.2 Faculty and Resource Persons

Name	Designation	Role in Programme
Prof. M.G. Cyriac	Associate Professor, Civil Engg., Jyothi Engineering College	Chief Supervisor; lectures on water quality, bacterial contamination, waste water, well pollution
Mrs. Jeffy Johny	Faculty, Jyothi Engineering College	Coordinator; lectures on water quality analysis and Kerala water issues; lab supervision
Mrs. Anna Joseph	Faculty, Jyothi Engineering College	Lectures on water treatment and sanitation; lab supervision

Name	Designation	Role in Programme
Mr. Suhas Nair	Faculty, Jyothi Engineering College	Coordinator; group supervisor (Groups 1 & 2); lab supervision
Mr. Ignatious	Faculty, Jyothi Engineering College	Lab supervision
Mr. Vini	Faculty, Jyothi Engineering College	Lab supervision
Dr. Anu Cyriac	Medical Officer, Health Department	Guest lecture: waterborne diseases, water-related diseases, Kerala scenario
Dr. Alwyn Varghese	Expert, Jyothi Engineering College	Industrial visit escort and guidance, Peechi Dam and KWA facilities

2. Day-by-Day Report of Programme Events

This section provides a detailed account of each day's events, activities, and observations as recorded by the Chief Supervisor.

Day 1

7 April 2026

Registration, Inauguration & Launch of Classroom Sessions

2.1 Registration and Inauguration

The programme commenced on the morning of April 7, 2026. Students from Markaz College of Arts and Science arrived at the CWRE premises and completed the registration formalities between 9:00 am and 9:30 am. The registration process was supervised by Mrs. Jeffy Johny and Mr. Suhas Nair. All 25 students enrolled for the programme were present, and attendance was recorded formally.

The inauguration ceremony was held from 9:30 am to 10:00 am. The event set a formal and motivated tone for the ten-day programme. The inaugural address emphasized the critical importance of water quality as a public health issue and the relevance of the internship to the students' professional development as microbiology graduates. The students and faculty were introduced, and the programme schedule was outlined.



Photograph 1 – Inauguration Ceremony, 7 April 2026

2.2 Introductory Session – Water Quality Scenario and Need for Analysis

Following the inauguration, I conducted the first substantive session from 10:00 am to 11:15 am. The session introduced students to the overarching context of the internship: the water quality scenario in India and Kerala, the concept of water pollution and its sources, and the public health rationale for systematic water quality analysis. Students were also briefed on the structure and expectations of the report they would be required to prepare collectively.

The session generated lively discussion. Students raised questions about specific water quality incidents they had observed in their home areas — including discolouration of well water, odour complaints, and suspected illness clusters linked to water consumption. This spontaneous engagement confirmed the relevance of the programme to the students' lived experience and set a participatory tone for subsequent sessions.

2.3 Session on Water Quality Analysis Methods

Mrs. Jeffy Johny conducted the second classroom session from 11:30 am to 12:30 pm. The session covered the range of methods employed in water quality analysis, with particular attention to volumetric and colorimetric analytical techniques, the principles underlying key instruments, and the water quality standards prescribed by the Bureau of Indian Standards (BIS) and the World Health Organisation (WHO). The eight parameters selected for the students' practical analysis exercises — pH, turbidity, conductivity, hardness, chloride, iron, BOD, and bacteriological indicators — were introduced in this session.

2.4 First Practical Analysis Session

The afternoon session (1:30 pm – 4:00 pm) marked the first practical hands-on exercise. Students were divided into five groups of five members each, as detailed in Section 4 of this report. Each group was assigned a dedicated laboratory bench and a set of water samples. The session was supervised jointly by Mrs. Vini, Mrs. Anna Joseph, Mr. Ignatious, and Mrs. Jeffy Johny. Students began familiarising themselves with laboratory equipment, reagents, and analytical procedures.



Photograph 2 – First Practical Analysis Session, Day 1

Day 2

8 April 2026

Bacterial Contamination, Water Treatment & Continued Analysis

2.5 Session on Bacterial Contamination of Water

Day 2 opened with a classroom lecture by me (9:30 am – 10:45 am) focused on the microbiology of water — a topic of direct relevance to the students' academic background in microbiology. The session covered the significance of bacterial contamination in water, the distinction between aerobic, anaerobic, and facultative bacteria, and the concept of indicator organisms. Particular emphasis was placed on the role of *E. coli* and the coliform group as indicators of fecal contamination and their significance in determining the safety of drinking water. The students engaged actively, drawing on their prior microbiology coursework, and several were able to contribute meaningfully to the discussion.

2.6 Session on Water Treatment

Mrs. Anna Joseph conducted the second session of the day (11:00 am – 12:30 pm) on water treatment — both at the conventional (treatment plant) scale and at the household/domestic level. The session described the sequential unit processes of a conventional treatment plant — coagulation and flocculation, sedimentation, filtration, and disinfection — and contrasted these with simpler, cost-effective domestic treatment methods including boiling, household chlorination, biosand filters, and solar disinfection. The session prepared students for the practical insights they would later gain during the industrial visit to the Peechi Water Treatment Plant.



Photograph 3 – Bacterial Analysis Discussion, Day 2

2.7 Continued Group Analysis Practice

The afternoon (1:30 pm – 4:00 pm) was devoted to a second round of group-based water quality analysis exercises. By Day 2, students had become more confident in handling equipment and following standard procedures. The supervisory team observed marked improvement in technique, teamwork, and speed compared to the previous day. Groups were encouraged to record their observations systematically in their field notebooks, forming the basis of their eventual reports.

Day 3

9 April 2026

Online Session on Well Pollution & Local Well Inspection

2.8 Online Session – Well Pollution and Remedies

Day 3 was structured differently from the preceding days, as it combined an online instructional session with independent field-based work by students in their own residential localities. I conducted an online session from 9:00 am covering the subject of well pollution in Kerala — its causes (proximity of sanitation facilities, poor construction standards, monsoon surface runoff), consequences for bacteriological water quality, and practical remediation measures including well disinfection using chlorination, structural repair of well walls and aprons, and the installation of sanitary well heads.

The session also introduced students to the methodology of well inspection — a structured visual and physical assessment of a well and its immediate surroundings to identify potential sources of contamination. Students were trained to use a simple inspection checklist covering the condition of the well wall, apron, drainage, cover, and the proximity and condition of nearby sanitation facilities.

2.9 Field Inspection of Wells in Students' Home Areas

Following the online session, students were deployed to their respective home areas to conduct practical well inspection exercises. Each student was tasked with identifying and inspecting at least one domestic water source — typically a dug well — in their neighbourhood, documenting observed sanitary deficiencies, recording GPS coordinates where possible, and gathering water samples for subsequent analysis. Students were required to prepare structured field notes and a preliminary report for submission.

This activity was designed to develop the students' capacity for independent field investigation and to ground their learning in the real-world water quality conditions of their own communities. The exercise also generated a valuable dataset of observations from multiple localities across the Malappuram region, which was later incorporated into the group reports and the awareness competition.

Day 4

10 April 2026

Kerala Water Quality Issues, Awareness Competition Launch & Analysis**2.10 Session on Water Quality Issues in Kerala**

Mrs. Jeffy Johny delivered a comprehensive session (9:30 am – 11:00 am) on the specific water quality challenges of Kerala State. The session addressed the prevalence of iron contamination in lateritic terrain, the high vulnerability of open dug wells to seasonal bacterial contamination, the emergence of fluoride issues in eastern districts, the problem of coastal salinization in the backwater belt, and the contribution of untreated domestic and industrial effluents to river and groundwater degradation. The Kerala Water Authority's response strategies — including source protection, piped water supply expansion, and community water quality monitoring — were also discussed.

Students from Malappuram district were particularly engaged with this session, as many could directly relate the content to conditions they had personally observed and documented during the previous day's field inspection exercise. The session reinforced the practical relevance of the internship's subject matter.

2.11 Launch of Water Quality and Sanitation Awareness Competition

I introduced the Water Quality and Sanitation Awareness Competition to the students between 11:15 am and 12:30 pm. The competition was designed to encourage students to synthesise their learning and communicate it creatively. Teams were briefed on the competition format, evaluation criteria (accuracy of content, clarity of communication, creativity, and presentation skills), and the timeline for submission and presentation on Day 9. The introduction of the competition generated considerable enthusiasm among the groups.



Photograph 5 – Session on Kerala Water Quality Issues, Day 4

2.12 Third Group Analysis Session

The afternoon (1:30 pm – 4:00 pm) was again devoted to group analysis exercises. By this stage, all groups were working efficiently and accurately on the physical and chemical parameters. Group supervisors observed and recorded each group's performance, which would contribute to the overall assessment of the programme.

Days 5 & 6

11 & 13 April
2026

Extended Field Visits – Local Area Water Source Survey

2.13 Field Survey – Day 5 (11 April 2026)

Day 5 was dedicated entirely to field visits, with students operating in their home localities from 9:00 am to 5:00 pm under the overall coordination of the programme coordinators. Building on the well inspection skills acquired on Day 3, students undertook a more extensive survey of domestic water sources in their respective areas. Each group focused on a cluster of households, systematically documenting the condition of water sources, sanitary surroundings, and any observable pollution hazards.

Water samples were collected from selected sources for later laboratory analysis. Students also engaged informally with household members to understand their water usage patterns, treatment practices (if any), and health experiences potentially linked to water quality. This community interaction dimension of the field work — though informal — proved to be a valuable learning experience, giving students insight into the gap between technical knowledge and community awareness on water quality issues.

2.14 Field Survey – Day 6 (13 April 2026)

Day 6 (April 13) continued the field survey programme, with students completing their data collection and field reporting tasks. Groups were required to compile their observations into a structured field report, covering the number and type of water sources surveyed, sanitary deficiencies identified, sample collection details, and preliminary field observations on water quality. The completed field reports were submitted to the group supervisors at the end of the day for review and feedback.

The two-day field programme was one of the most distinctive and educationally valuable components of this internship. It is relatively rare for college programmes to provide students with extended, independent field exposure of this kind, and the quality of the field reports submitted at the end of Day 6 reflected the seriousness with which students had engaged with the exercise.

Day 7

16 April 2026

Report Writing Guidance, Guest Lecture on Waterborne Diseases & Analysis**2.15 Report Writing Discussion**

The programme resumed on April 16 after the two-day field phase. Mrs. Anna Joseph and Mrs. Jeffy Johny conducted a focused session on report writing (9:30 am – 10:30 am). Field reports submitted by the groups on Day 6 were reviewed collectively. Common shortcomings — including inconsistent data recording, lack of systematic structure, and the need for clear cross-referencing between observations and supporting data — were identified and addressed. Guidance was provided on the expected format and content of the final internship report, including the chapter structure, appropriate use of tables and figures, citation practices, and certification requirements.

2.16 Guest Lecture – Dr. Anu Cyriac, Medical Officer

A highlight of Day 7 was the guest lecture delivered by Dr. Anu Cyriac, Medical Officer with the Kerala Health Department, from 10:45 am to 12:30 pm. Dr. Cyriac's lecture brought a clinical and epidemiological perspective to the programme, complementing the technical water quality content covered in preceding sessions. The lecture addressed the classification of water-related diseases — waterborne, water-washed, water-based, and water-related vector-borne diseases — with specific reference to the Kerala scenario.

Dr. Cyriac described in detail the epidemiology of common waterborne diseases including cholera, typhoid, hepatitis A, and dysentery, and the water-vector-borne diseases such as dengue, chikungunya, and malaria that continue to be a significant public health burden in Kerala. The discussion of the 2018 and subsequent flood events and the associated disease outbreaks was particularly relevant and resonated strongly with the students. The lecture stimulated extensive questioning and discussion, and was unanimously acknowledged by participants as one of the most engaging sessions of the programme.



Photograph 8 – Guest Lecture by Dr. Anu Cyriac, Medical Officer, Day 7

2.17 Fourth Group Analysis Session

The afternoon session continued with group-based water quality analysis (1:30 pm – 4:00 pm). By Day 7, students had completed the bulk of their physical and chemical analyses. Groups began cross-checking and tabulating their accumulated results in preparation for the final presentation. The supervisory team provided feedback on analytical technique and data interpretation.

Day 8

17 April 2026

Industrial Visit – Peechi Dam, KWA Water Treatment Plant & KWA Laboratory

2.18 Industrial Visit – Overview

Day 8 was devoted entirely to the industrial visit — undoubtedly the most experientially rich day of the programme. Under the guidance of Dr. Alwyn Varghese and myself, accompanied by Mrs. Anna Joseph and Mrs. Jeffy Johny, all 25 students visited the Peechi Dam, the Kerala Water Authority (KWA) Water Treatment Plant at Peechi, and the KWA Water Testing Laboratory. The visit ran from 9:30 am to 4:00 pm.



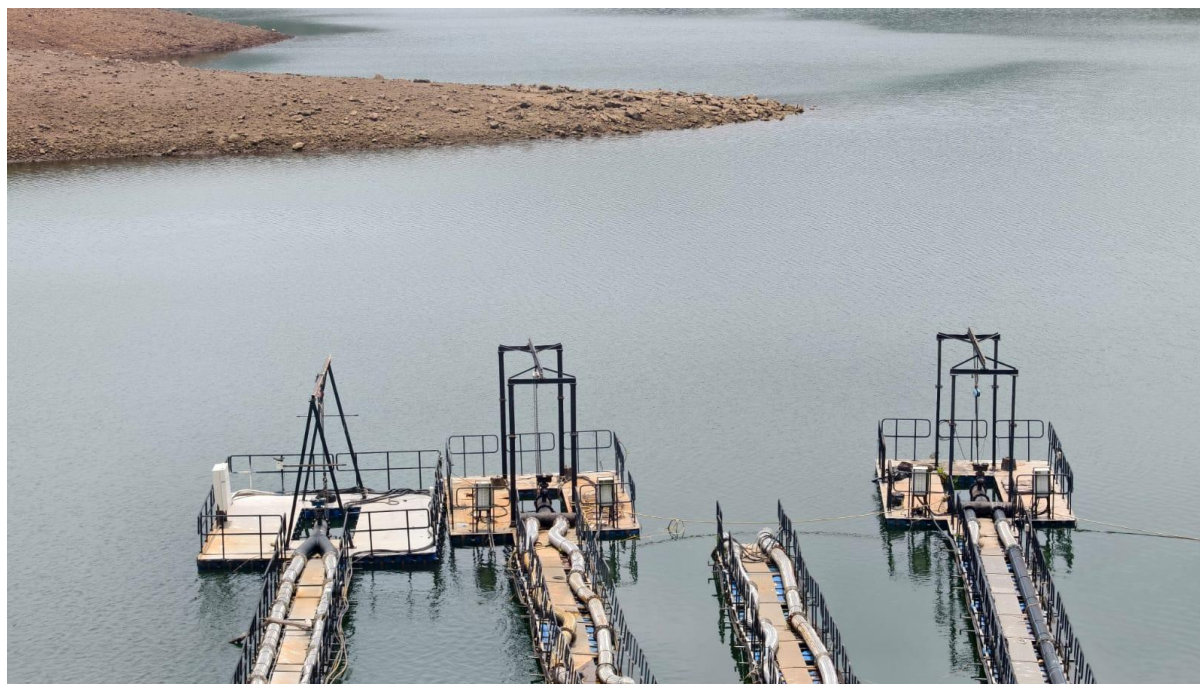
Photograph 9 – Group at Peechi Dam, Day 8 Industrial Visit

2.19 Peechi Dam

The visit began at the Peechi Dam, a major reservoir on the Manali River that serves as the primary raw water source for the KWA treatment plant. Students were briefed on the catchment area, storage capacity, and the hydrological and seasonal factors that influence raw water quality. The dam site provided a vivid illustration of the relationship between watershed land use — including agricultural activity and forest cover — and the quality of the raw water that must subsequently be treated.

Of particular interest to the students was the floating well structure at the reservoir, which is used by KWA to draw water from a consistent depth, avoiding the turbidity and temperature extremes of surface water and the dissolved solids concentration of deep bottom water. The

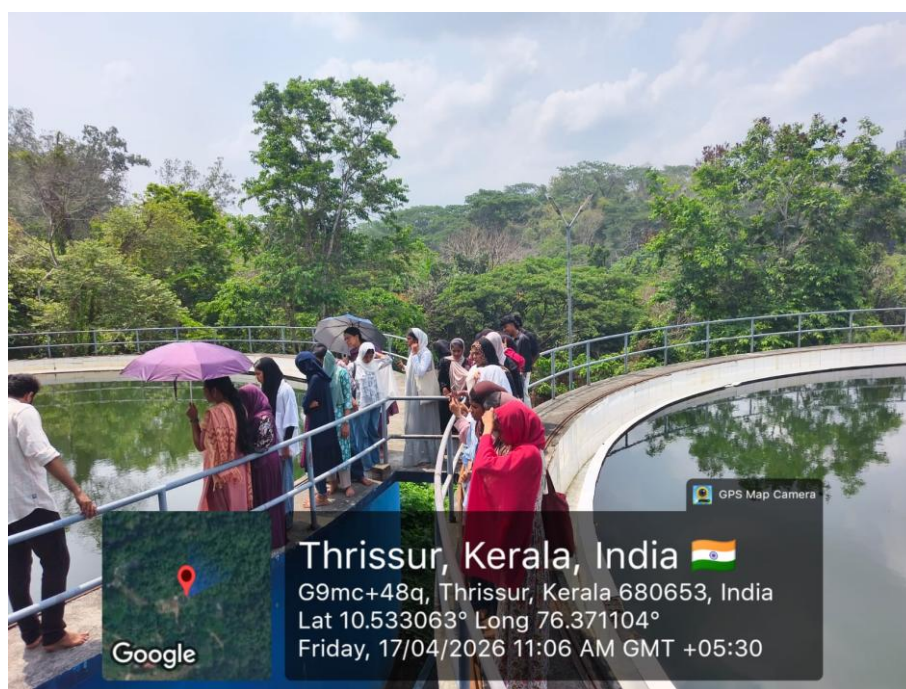
floating well concept was new to most students and generated considerable curiosity and discussion.



Photograph 10 – KWA Floating Well at Peechi Reservoir

2.20 KWA Water Treatment Plant

The group then toured the KWA Water Treatment Plant, where students were able to observe all stages of the conventional treatment process in operation — coagulation and flocculation chambers, sedimentation tanks, rapid sand filters, and the chlorination facility. Plant engineers provided commentary on the operational parameters, chemical dosing practices, and quality control procedures. The contrast between the turbid raw water entering the plant and the clear, treated water at the outlet made a powerful impression on students.



Photograph 11 – Students at the KWA Water Treatment Plant Facility

Several students drew direct connections between the theoretical content covered in Mrs. Anna Joseph's session on Day 2 and the operational reality they were witnessing at the plant. This integration of classroom learning and field observation is precisely the pedagogical outcome the industrial visit was designed to achieve.

2.21 KWA Water Testing Laboratory

The visit concluded at the KWA Water Testing Laboratory, where senior laboratory staff demonstrated the range of physical, chemical, and bacteriological tests routinely performed on water samples drawn from the distribution network. Students observed professional laboratory practice, quality assurance protocols, and the documentation systems used to record and report test results. The bacteriological testing procedures — membrane filtration and incubation on selective media — were of particular interest given the students' microbiology background. Several students noted the parallels and differences between KWA's professional testing protocols and the exercises they had been conducting themselves during the internship.

Day 9

18 April 2026

Sanitation of Buildings, Awareness Competition & Final Analysis

2.22 Session on Sanitary Measures for Buildings and Surroundings

Mrs. Anna Joseph conducted a session (9:30 am – 11:00 am) on the application of sanitation principles to the built environment. The session covered the spatial planning of water sources and sanitation facilities in residential settings, standards for well construction and protection (including parapet height, concrete apron, drainage diversion, and cover requirements), domestic wastewater management through properly designed septic systems, and the importance of solid waste management in preventing secondary contamination of water sources. Students were encouraged to reflect on the sanitary conditions they had observed during their field visits and to apply these standards retrospectively to evaluate their findings.

2.23 Water Quality and Sanitation Awareness Competition

The Water Quality and Sanitation Awareness Competition was held between 11:00 am and 12:30 pm on Day 9. This competition, which had been introduced to the students on Day 4, invited each group to present their findings, insights, and awareness messages on water quality and sanitation in a creative and communicative format of their choosing. The presentations were evaluated by a panel that I chaired, with the coordinators serving as assessors.

The quality and diversity of the presentations exceeded our expectations. Groups employed a variety of formats including informational posters, demonstration setups using actual water samples, role-play scenarios depicting community interactions about water quality, and slide-based presentations combining their own field data with Kerala-specific statistics. The student engagement and creativity on display were a testament to the success of the experiential learning approach adopted throughout the programme.

Group 3, supervised by Mrs. Anna Joseph, achieved the highest presentation score of 19 out of 20. Groups 1 and 2 (supervised by Mr. Suhas Nair) each scored 18, while Groups 4 and 5 (supervised by Mrs. Jeffy Johny) each scored 17. All groups performed creditably.



Photograph 13 – Water Quality and Sanitation Awareness Competition, Day 9

2.24 Final Group Analysis Session

The afternoon (1:30 pm – 4:00 pm) was used for the final round of laboratory analysis, with groups completing the bacteriological testing of their water samples — the most technically demanding component of the analysis programme. The supervisory team provided intensive support during this session, ensuring that students followed correct aseptic technique and properly documented their results. By the end of the afternoon, all groups had a complete dataset covering all eight water quality parameters, ready for incorporation into their final reports.

Day 10
20 April 2026

Wastewater Treatment, Group Presentations & Valedictory

2.25 Session on Introduction to Wastewater Treatment

The final day of the programme began with a session I delivered (9:30 am – 10:45 am) introducing the principles and processes of wastewater treatment. The session described the distinction between greywater and blackwater, the stages of wastewater treatment (preliminary, primary, secondary, and tertiary), the principles of biological treatment (activated sludge, trickling filters), and the importance of proper sludge management. The session also touched on the emerging relevance of water reuse as a sustainable approach to water resource management. This topic, while not directly tied to the water quality analysis exercises, was included to give students a complete picture of the water-sanitation cycle from source to disposal.

2.26 Final Group Presentations

The group presentations (11:00 am – 12:30 pm) were the culminating academic event of the internship. Each of the five groups presented their complete internship findings — encompassing the outcomes of their water quality analyses, observations from the field surveys, key learning from the industrial visit, and their conclusions regarding water quality conditions in their study areas. Presentations were evaluated on the quality of the analysis, logical organisation, use of data, and presentation skills.

The presentations were of a high standard, demonstrating that students had synthesised the diverse inputs of the programme — lectures, laboratory work, field observations, guest sessions, and the industrial visit — into coherent, evidence-based assessments. The panel was particularly impressed by the groups' ability to contextualise their laboratory findings within the broader framework of Kerala's water quality challenges.



Photograph 14 – Group Presentations, Day 10

2.27 Valedictory Function and Certificate Distribution

The programme concluded with the valedictory function held from 1:30 pm to 3:00 pm. The event provided an opportunity to formally celebrate the students' achievement and to reflect on the overall experience of the ten-day programme. Certificates of successful completion were distributed to all 25 participants. The certificates, countersigned by myself as Chief Supervisor, attest to each student's completion of more than 60 hours of intensive internship activities.

Students and faculty had the opportunity to share their reflections on the programme. The feedback from students was overwhelmingly positive, with many describing the internship — particularly the field visits and the industrial visit to Peechi — as among the most impactful learning experiences of their undergraduate education. Several expressed a wish that the programme could be extended, and a number indicated that the experience had influenced their thinking about career directions in environmental science and public health.

I expressed my appreciation to the students for their outstanding conduct and commitment throughout the programme, and to my colleagues Mrs. Jeffy Johny, Mrs. Anna Joseph, Mr. Suhas Nair, Mr. Ignatious, Mr. Vini, Dr. Anu Cyriac, and Dr. Alwyn Varghese for their dedicated and expert contributions.



Photograph 15 – Valedictory Ceremony and Certificate Distribution, 20 April 2026

3. Student Groups and Supervisors

Students were organised into five groups of five members each at the start of the programme. This grouping structure was maintained throughout all practical, field, and presentation activities.

Group	Members	Group Supervisor
1	Fathima Bathul, Jumana, Rinsa, Anshida Farvin A., Faseeha Nazrin T.	Mr. Suhas Nair
2	Fathima Bareera, Sneha Santhosh, Shafana Sharin, Shifana K., Minha	Mr. Suhas Nair
3	Fathima Misna, Fathima Minha P.V., Krishnaja P., Fidha, Shafeequa P.V.	Mrs. Anna Joseph
4	Ayasha Hena, Frashana Sherin, Muhammed Ramees P.T., Shumoocha, Ameer Sadath	Mrs. Jeffy Johny
5	Ayasha Hena, Frashana Sherin, Muhammed Ramees P.T., Shumoocha, Ameer Sadath	Mrs. Jeffy Johny

4. Student Performance Assessment

4.1 Written Test

A written test of 60 marks was administered to assess individual learning outcomes. The test covered water quality parameters, analytical methods, microbiology of water, Kerala water quality issues, and treatment processes. The results are recorded below.

Sl.	Name	Marks (out of 60)	Remarks
1	Asna P.	56	First Prize
2	Shafeeqa P.V.	55	Second Prize
3	Shumoooha	55	Second Prize
4	Anshida	53	
5	Krishnaja P.	53	
6	Bareena U.	–	
7	Fathima Rifa M.	52	
8	Fathima Thesni	51	
9	Fathima Bathal	51	
10	Minha M.	51	
11	Faseeha Nasrin T.	51	
12	Fathima Rinsa	50	
13	Ashila Farwin	50	
14	Shafna Sharin K.	49	
15	Jumana K.	49	
16	Fathima Aiban B.	49	
17	Fathima Rinsha P.V.	–	
18	Fathima Rinsa	48	
19	Aysha Hena	48	
20	Farsana Sherin P.V.	48	
21	Fidha C.	47	
22	Shifna K.	47	
23	Sneha Santhosh	44	
24	Ameer Sadath	41	
25	Muhammed Ramees	40	

The performance in the written test was commendable. The top score of 56 out of 60 was achieved by Asna P. The majority of students scored above 47, reflecting a solid grasp of the programme content. The results demonstrate effective knowledge transfer across the range of topics covered during the ten-day programme.

4.2 Group Presentation Scores

Group	Members	Supervisor	Score /20
1	Fathima Bathul, Jumana, Rinsa, Anshida Farvin, Faseeha Nazrin	Mr. Suhas Nair	18
2	Fathima Bareera, Sneha Santhosh, Shafana Sharin, Shifana K., Minha	Mr. Suhas Nair	18
3	Fathima Misna, Fathima Minha P.V., Krishnaja P., Fidha, Shafeequa P.V.	Mrs. Anna Joseph	19
4	Ayasha Hena, Frashana Sherin, Muhammed Ramees P.T., Shumoocha, Ameer Sadath	Mrs. Jeffy Johny	17
5	Ayasha Hena, Frashana Sherin, Muhammed Ramees P.T., Shumoocha, Ameer Sadath	Mrs. Jeffy Johny	17

4.3 General Assessment and Observations

Beyond the formal test and presentation scores, I wish to place on record the following observations on the students' overall performance and conduct during the programme:

- **Attendance and Punctuality:** Attendance was near-perfect across all ten days. Students were punctual for sessions and demonstrated strong commitment to the programme schedule.
- **Laboratory Discipline:** Students observed all laboratory safety and hygiene protocols consistently. Their handling of analytical equipment showed steady improvement from Day 1 to Day 9.
- **Field Work Quality:** The quality of field reports submitted after Days 3, 5, and 6 was impressive, with students demonstrating systematic observation skills and attention to detail.
- **Teamwork:** All five groups functioned cohesively, with group supervisors consistently reporting positive inter-personal dynamics and equitable distribution of work.
- **Engagement with Resource Persons:** Students showed respectful and intellectually engaged interaction with all faculty and guest lecturers. The discussion generated during Dr. Anu Cyriac's guest lecture in particular was of a quality one would expect from final-year postgraduate rather than undergraduate students.
- **Conduct:** Without exception, students maintained a high standard of personal conduct and decorum throughout the programme, including during the industrial visit and field excursions. This was specifically noted and appreciated by KWA officials at Peechi.

5. Chief Supervisor's Overall Assessment

It is my considered assessment that the internship programme on Water Quality and Sanitation conducted at the Centre for Water Research and Education, Jyothi Engineering College, during April 2026 was a thorough and highly successful programme in all respects.

The programme was carefully designed to provide a multi-modal learning experience: classroom instruction delivered by qualified faculty, hands-on laboratory practice, independent field investigation, engagement with a senior public health professional, a comprehensive industrial visit, and group presentation and report writing. Each component was deliberately chosen to build on and reinforce the others, creating a coherent and integrated learning arc.

The 25 students from Markaz College of Arts and Science engaged with every component of the programme with a level of seriousness, curiosity, and enthusiasm that made the task of supervising the programme genuinely rewarding. Their academic background in microbiology proved to be an excellent foundation for the interdisciplinary content of the programme, and many students were able to make connections between their prior learning and the new material that enriched the discussions for the entire group.

The programme fully met its stated objectives. All 25 students successfully completed more than 60 hours of programme activities, met the requirements for the written test, participated in the group analysis exercises, completed field reports, and delivered group presentations. Certificates have been issued to all students accordingly.

I would like to formally recommend this internship model — or an expanded version of it — for adoption by other colleges and institutions seeking to provide their students with meaningful, experience-based learning in the fields of water quality, environmental science, and public health. The collaboration between Markaz College of Arts and Science and the CWRE at Jyothi Engineering College has been a productive and mutually beneficial one, and I would welcome the opportunity to continue and expand this partnership.

6. Certificate of Completion

CERTIFICATE

This is to certify that 25 students from Markaz College of Arts and Science, Athavanad have attended an Internship on Water Quality and Sanitation during the days 7, 8, 9, 10, 11, 13, 16, 17, 18, and 20 of April 2026 at the Centre for Water Research and Education (CWRE), Jyothi Engineering College, Cheruthuruthy.

The internship was held in a very conducive atmosphere and the students maintained utmost decency and dedication in all aspects, which was appreciated by all concerned. They have completed more than 60 hours of work very effectively.

The programme was conducted in accordance with its stated objectives and the prescribed schedule. All students are hereby certified as having successfully completed the programme.

Place: Cheruthuruthy

Date: April 2026

Prof. M.G. Cyriac

Chief Supervisor

Centre for Water Research and Education (CWRE)
Jyothi Engineering College, Cheruthuruthy

Appendix: Biodata of the Chief Supervisor

Sl.	Qualification / Role	Institution / Details
Academic Qualifications		
1	B.Sc. Engg. Civil, 1981	NIT Calicut
2	M.Tech. Environmental Engineering, 1994	Sree Jayarama Rajendra College of Engineering, Mysore
3	Management Development Diploma, 2005	IIM Bangalore
Professional Experience		
1	Director, Communication & Capacity Development Unit, WRD Kerala (2010–2012, deputation)	Government of Kerala, Water Resources Department
2	Asst. Engineer to Executive Engineer (1982–2013)	Kerala Water Authority
3	Manager Environment, Kerala Rural Water Supply and Sanitation (2010–2012)	KRWSA
4	Environmental Safeguard Expert, Kerala Solid Waste Management Project (2022–2023)	Government of Kerala
5	Associate Professor, Civil Engineering (2013 – continuing)	Jyothi Engineering College, Cheruthuruthy

– End of Report –